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Modelling and simulating elephant movement with step selection functions

GPS data from bull elephants in Botswana

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1 Introduction

This master's thesis is carried out in collaboration with Elephants for Africa, a research and conservation organisation in Botswana. It is supervised by Jonas Skeppstedt, Department of Computer Science, and Johan Lindström, Mathematical Statistics, both at Lund University. The work is planned to take place between February 2027 and June 2027. This document describes the background, goals, research questions and approach for a master's thesis carried out at Lund University, Department of Computer Science.

2 Background and motivation

Botswana has the world's largest elephant population and as elephants return to areas they have been absent from for decades, they increasingly come into conflict with people. Being able to anticipate where elephants will move, and what draws them there, would give communities and conservation organisations time to prepare for such movement. For computer science students the problem is also interesting from a modelling perspective, where it combines large amounts of GPS data, spatial environmental data, statistical models and newer machine learning methods and applies knowledge from several courses in the program to a real problem.

2.1 Human–elephant conflict in Botswana

Botswana has around 130,000 elephants, and their range has expanded by more than 50 percent in 20 years, most notably south towards the Makgadikgadi [1]. Between 2018 and 2022, about 61 people were killed and 54 injured in conflicts with wildlife, and elephants caused 41 percent of the damage to crops, boreholes, water tanks and fences [2]. The conflict also harms elephants, when they are killed in retaliation after an incident [3].

In the Makgadikgadi, the Boteti River began flowing again in 2009 after almost 20 dry years. Elephants were present before that, but not in such large numbers, and since then they have increasingly returned to their historical range [4]. Many communities near the national park have not lived with elephants for a generation, and the knowledge of how to do so has been lost [4]. The male elephants in the area spend about three times as much time outside protected areas as inside them [5], but where they go and why, is still not well understood [4].

2.2 What drives elephant movements?

Water is the best known driver, and new boreholes draw elephants into areas where they were previously absent [6]. Minerals such as sodium, calcium and phosphorus are less studied. Where plants and water contain too little sodium, elephants seek out and eat salty soil [7], and a recent study warns that animals travelling far for salt may come into more frequent conflict with people [8]. Since the Makgadikgadi is a landscape of salt pans, the question is especially relevant there.

2.3 Step selection functions

A step selection function (SSF) is a standard statistical method for analysing animal movement from GPS data [9]. Each observed step, the move between two consecutive positions, is compared with a set of random steps the animal could have taken from the same starting point. The model then learns which environmental conditions, such as distance to water or type of vegetation, make a step more likely to be chosen. Integrated SSFs (iSSF) estimate the movement itself, such as step length and turning angle, together with habitat selection in one model [10].

A fitted SSF can also be used to simulate new tracks. By simulating many tracks forward in time, one obtains a probability map of where the animal is likely to be, rather than a single route [11]. This is what makes SSFs interesting for predicting elephant movements over longer periods. The simulated tracks also show corridors where movement is concentrated, and SSFs have previously been used to map corridors for elephants in Kenya and Tanzania [12].

2.4 Deep learning step selection functions

A standard SSF is a log-linear model, so non-linear effects and interactions, for example that salt pans only attract elephants in the dry season, must be specified in advance. Recent work replaces the linear model with a neural network. deepSSF takes environmental maps as input and outputs a probability map for the next step, learning features such as rivers and forest edges without them being specified [13]. On GPS data from water buffalo in northern Australia, it generally predicted better than a corresponding iSSF [13].

2.5 The data

Elephants for Africa has tracked around ten bull elephants in the Makgadikgadi region with GPS collars over several years [14]. The exact period and sampling rate will be confirmed with Elephants for Africa. Soil samples have been collected [15], but they are still awaiting export for analysis and the results will probably not be available for the thesis. Mineral data is therefore compiled through a literature review of soil and geological maps, published studies and data from the mining industry. Before the thesis starts, the data will be prepared so that the work can move quickly once on site, and environmental data such as water, vegetation and salt pans will be added from open satellite sources and on-site mappings.

2.6 Elephants for Africa's part in the work

Elephants for Africa is a research and conservation organisation that has studied male elephants since 2012 [4]. It wants to anticipate where the elephants will move over seasons and years, so that it can inform villages and plan measures such as new salt licks or changes to waterholes. One of the original aims of its GPS project was to find out whether mineral needs drive the bulls' movements [15], which this thesis addresses. Elephants for Africa provides the GPS data and the knowledge about the elephants. The contact person is

Kate Evans, founder and director of research at Elephants for Africa, who is also affiliated with the University of Gothenburg. The thesis has no formal connection to the University of Gothenburg.

3 Research questions and goals

The thesis aims to investigate whether step selection functions can be used to predict the movements of bull elephants in the Makgadikgadi, and whether minerals influence where they move. Three models of increasing complexity are compared: a classical SSF, an integrated SSF (iSSF) and a deep learning SSF (deepSSF). The models are used to simulate where the elephants are likely to be 1, 3, 6 and 12 months ahead, and the resulting probability maps are compared with the elephants' actual positions. Water and vegetation are well-known drivers of elephant movement and are included in all models, while the focus is on the less studied role of minerals. The following research questions will be investigated:

> **RQ1:** How well do an SSF, an iSSF and a deepSSF predict the movements of bull elephants 1, 3, 6 and 12 months ahead, for elephants and time periods not used for training, compared with a seasonal baseline based on the elephants' previous positions? Does added model complexity improve the predictions, and if so, over which time horizons?
>
> **RQ2:** Do minerals influence where the elephants move? When water and vegetation are accounted for, does adding mineral sources such as salt pans and mineral maps from the literature improve the models, and does the effect vary between seasons?
>
> **RQ3:** Are there corridors or bottlenecks where many elephants pass, and do they coincide with mineral-rich areas?

All three questions can be answered with the GPS data, open environmental data and mineral data from the literature. If time allows, a small number of soil samples may be taken along the corridors identified in RQ3 to check the mineral maps from the literature. The thesis does not depend on them. A further goal is to deliver seasonal probability maps that Elephants for Africa can use when informing villages and planning measures.

4 Approach

To structure the thesis, it is divided into three phases:

1. **Discovery phase**: Literature study of step selection functions, deep learning for animal movement and what drives elephant movement. The GPS data is cleaned and resampled to a suitable step length, and environmental layers for water, vegetation and salt pans are built from open satellite data. Layers for soil minerals are built from open soil maps and a literature review of soil and geological maps, publications and mining-industry data.

2. **Implementation and test phase**: An SSF, an iSSF and a deepSSF are implemented in Python and trained on the GPS data. The fitted models are used to simulate many tracks 1, 3, 6 and 12 months ahead. This gives probability maps of where the elephants are likely to be, and shows corridors where many tracks pass.

3. **Evaluation phase**: The models are tested on data they have not seen. Each elephant is held out in turn, and the last year of data is used as the test period. The predictions are compared with the elephants' actual positions and with a simple baseline map of where the elephants have been in the same season before. The effect of minerals is assessed by comparing models with and without the mineral layers. The results are then discussed and summarised in a conclusion, and the probability maps are delivered to Elephants for Africa.

To make sure the project progresses as planned, the following checkpoints are set:

- **End of the discovery phase:** Review of the literature and the prepared data, and planning of the implementation.
- **Halfway through the implementation phase:** Review of the first working model (the SSF) and any adjustments.
- **After the implementation and test phase:** Review of the results from all three models.
- **Final checkpoint:** Presentation of the conclusions and review of the final report before submission.

5 Workplace and resources

The thesis is carried out partly in Botswana, at Elephants for Africa's research camp in the Makgadikgadi, and partly in Sweden. Elephants for Africa provides the GPS data and knowledge about the elephants and the area. Mineral data is gathered through a literature review of soil and geological maps, publications and mining-industry data, together with open soil maps and satellite data. Supervision is given by Jonas Skeppstedt and Johan Lindström at Lund University. Support on the elephants and the area is given by Kate Evans at Elephants for Africa, and by a local supervisor in Botswana who is still to be confirmed.

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